

The absolute Jacobian integral of a radial sphere extension

MathlibAnnex.Sphere.radialExtension_integral_abs_det

theorem

Identifies the integral of the absolute Jacobian over an open source unit ball with the volume of the closed target unit ball.

Statement

Let $m \in \mathbb{N}_0$, put $n = m + 1$, and let $E = \mathbb{R}^n$ have its standard sup norm $\|\cdot\|_\infty$ and Lebesgue measure λ . Let p, q be continuous seminorms on E with specified positive constants a_p, b_p, a_q, b_q satisfying

$$a_p\|x\|_\infty \leq p(x) \leq b_p\|x\|_\infty, \quad a_q\|x\|_\infty \leq q(x) \leq b_q\|x\|_\infty \quad (x \in E).$$

The lower bounds make p and q norms. Define their unit spheres by $\Sigma_p = \{x : p(x) = 1\}$ and $\Sigma_q = \{y : q(y) = 1\}$. Let $\Delta : \Sigma_p \rightarrow \Sigma_q$ be a bijection satisfying

$$q(\Delta(u) - \Delta(v)) = p(u - v) \quad (u, v \in \Sigma_p).$$

Its radial extension, read as a map $R_\Delta : E \rightarrow E$ in the fixed coordinates, is

$$R_\Delta(x) = \begin{cases} 0, & x = 0, \\ p(x) \Delta(x/p(x)), & x \neq 0. \end{cases}$$

For $x \neq 0$, the normalized vector $x/p(x)$ belongs to Σ_p , so this expression is defined. Put

$$B_p = \{x : p(x) < 1\}, \quad \overline{B}_q = \{y : q(y) \leq 1\}.$$

At a differentiability point x , let $DR_\Delta(x) : E \rightarrow E$ be the Fréchet derivative and set

$$j_\Delta(x) = |\det DR_\Delta(x)|.$$

At other points set $j_\Delta(x) = 0$. Thus j_Δ is the absolute Jacobian, not a norm of $R_\Delta(x)$ or a determinant of the nonlinear map itself. Then

$$\int_{B_p} j_\Delta(x) d\lambda(x) = \lambda(\overline{B}_q),$$

where the finite measure on the right is identified with its real value.

Assumptions

The dimension is the positive integer $n = m + 1$. The maps defining p and q are continuous seminorms with the displayed positive comparison bounds. The sphere map Δ is a bijective isometry for the distances induced by p and q .

The derivative and the measure in the integral use the fixed coordinate space $(E, \|\cdot\|_\infty)$, whereas the unit spheres use p and q . No linearity or everywhere differentiability of R_Δ is assumed.

Conclusion

The left side is the integral of the nonnegative function j_Δ over the open p -unit ball. At a differentiability point, with coordinate vectors $e_1, \dots, e_n$ and component functions $R_{\Delta,i}(x) = (R_\Delta(x))_i$, the integrand is explicitly

$$j_\Delta(x) = |\det(\partial_j R_{\Delta,i}(x))_{1 \leq i,j \leq n}|,$$

$$\partial_j R_{\Delta,i}(x) = (DR_\Delta(x)e_j)_i.$$

The right side is the volume of the closed q -unit ball in the same coordinates. The theorem uses the absolute Jacobian and does not assert that the radial extension is linear.

Changing from the norm p or q to the sup norm changes the normed-space structure, not the coordinate vector. The reference Lebesgue normalization remains fixed. At a point without a derivative, the source takes the zero linear map; since $n = m + 1 > 0$, its determinant is zero, exactly matching the definition of j_Δ above.

Proof route

Prove global Lipschitz control and the exact open-ball image. Apply the nonnegative area formula after removing the null nondifferentiability set, justify the real integral, and use the zero-measure boundary of the target ball.

▼ Proof steps

1. **Control the map and identify its image.** Radius preservation and the inverse radial extension give

$$q(R_\Delta(x)) = p(x), \quad R_{\Delta^{-1}}(R_\Delta(x)) = x \quad (x \in E).$$

In particular, R_Δ is injective. The radial three-Lipschitz estimate and the comparison bounds give

$$\begin{aligned} q(R_\Delta(x) - R_\Delta(x')) &\leq 3p(x - x'), \\ \|R_\Delta(x) - R_\Delta(x')\|_\infty &\leq \frac{3b_p}{a_q} \|x - x'\|_\infty. \end{aligned}$$

This is the reference-coordinate Lipschitz bound for the same p, q, Δ . Define the open target ball $B_q = \{y : q(y) < 1\}$. The radius identity gives $R_\Delta(B_p) \subseteq B_q$. Conversely, for $y \in B_q$, take $x = R_{\Delta^{-1}}(y)$; then $p(x) = q(y) < 1$ and $R_\Delta(x) = y$. Hence

$$R_\Delta(B_p) = B_q,$$

as stated by the exact open-ball image theorem.

2. **Remove null complements on the source and target.** Set

$$G = B_p \cap \{x : R_\Delta \text{ is differentiable at } x\}, \quad N = B_p \setminus G.$$

Then

$$\lambda(N) = 0, \quad \lambda(R_\Delta(N)) = 0.$$

The first is Rademacher's theorem applied to the Lipschitz bound of Step 1. The second is the equal-dimensional Lipschitz null-image theorem for that same map and the null set N ; it is recorded in the radial image-exception lemma. The set G is measurable. Since R_Δ is injective and $R_\Delta(B_p) = B_q$,

$$B_q \setminus R_\Delta(G) = R_\Delta(N).$$

Thus the omitted target subset is null and $\lambda(R_\Delta(G)) = \lambda(B_q)$.

3. **Apply the nonnegative area formula.** Write $\int^+$ for the nonnegative extended integral, valued in $[0, \infty]$. The formula is

$$\int_G^+ j_\Delta(x) d\lambda(x) = \lambda(R_\Delta(G)).$$

Apply the injective nonnegative area formula with measurable domain G , map R_Δ , derivative $DR_\Delta(x)$ at each $x \in G$, injectivity from Step 1, and reference Lebesgue measure λ . Step 2 then yields

$$\int_{B_p}^+ j_\Delta d\lambda = \int_G^+ j_\Delta d\lambda = \lambda(R_\Delta(G)) = \lambda(B_q).$$

The first equality uses $\lambda(B_p \setminus G) = 0$. These are extended nonnegative integrals; real integrability is established next.

4. **Verify integrability of the same integrand.** Put $\overline{B}_p = \{x : p(x) \leq 1\}$. This is compact by the comparison bounds. The integrability estimate is

$$\int_{\overline{B}_p} j_\Delta(x) d\lambda(x) = \int_{\overline{B}_p} |\det DR_\Delta(x)| d\lambda(x) < \infty.$$

For compact-set integrability of Lipschitz derivative minors, use R_Δ , its reference-coordinate Lipschitz bound, the compact set $\overline{B}_p$, and all $n = m + 1$ coordinates in increasing order. The resulting minor is $\det DR_\Delta$; taking the absolute value gives j_Δ . This application is recorded by integrability of the radial absolute determinant. Since $B_p \subseteq \overline{B}_p$, we have $j_\Delta \in L^1(B_p, \lambda)$.

Because $j_\Delta \geq 0$, its finite extended integral is the same number as its real integral. Taking real values in Step 3 therefore gives

$$\int_{B_p} j_\Delta(x) d\lambda(x) = \lambda(B_q),$$

with the finite measure on the right read as a real number.

5. **Replace the open target ball by the closed ball.** The boundary identities are

$$\partial B_q = \{y : q(y) = 1\}, \quad \lambda(\partial B_q) = 0,$$

$$\overline{B}_q = B_q \cup \partial B_q, \quad \lambda(\overline{B}_q) = \lambda(B_q) < \infty.$$

The boundary description follows from the norm-ball description; its measure is zero by the convex-set boundary theorem. Both steps are used in the open/closed target-volume lemma. Finiteness follows from compactness of the closed q -unit ball. Substitution into Step 4 gives the stated equality

$$\int_{B_p} j_{\Delta}(x) d\lambda(x) = \lambda(\overline{B}_q).$$

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.radialExtension_integral_abs_det`
- Reference-coordinate Lipschitz control — `MathlibAnnex.Sphere.lipschitzWith_radialMap`
- Null images in equal dimension —
`MathlibAnnex.BilipschitzOrientation.volume_image_eq_zero_of_lipschitzWith`
- The radial image exception is null — `MathlibAnnex.Sphere.radial_image_badSet_null`
- Nonnegative injective absolute area formula —
`MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image`
- Compact-set integrability of derivative minors —
`MathlibAnnex.NullLagrangian.integrableOn_maximalMinor_fderiv_of_lipschitzWith`
- Integrability of the radial absolute determinant —
`MathlibAnnex.Sphere.integrableOn_radial_abs_det`
- Open and closed target ball volumes agree —
`MathlibAnnex.Sphere.modelOpenBall_volume_eq_ballVolume`
- The radial extension maps open balls exactly — `MathlibAnnex.Sphere.radialExtension_image_ball`
- Rademacher for globally Lipschitz maps in finite dimension — `LipschitzWith.ae_differentiableAt`

Lean source signature (exact)

```
theorem radialExtension_integral_abs_det {m : ℕ}
  {MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)}
  (Δ : sphere (0 : Space MX) 1 ≈i sphere (0 : Space MY) 1) :
  ∫ x in MX.p.ball 0 1,
    |ContinuousLinearMap.det (fderiv ℝ (fun y : Fin (m + 1) → ℝ =>
      (show Fin (m + 1) → ℝ from radialExtension (X := Space MX) (Y := Space MY) Δ (show
Space MX from y))) x)|
    ∂volume = MY.closedUnitBallVolume
```

In the source	Mathematical meaning
<code>{m : ℕ} {MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)}</code>	The positive dimension $n = m + 1$ and the two stored seminorms p, q on the fixed sup-norm coordinate space $E = \mathbb{R}^n$, with the positive comparison bounds stated in this Card.
<code>Space MX; Space MY</code>	The same coordinate vectors equipped with norms p and q . This changes the norm used for sphere distances, not the reference coordinate measure.

In the source	Mathematical meaning
$(\Delta : \text{sphere } (\theta : \text{Space MX}) \ 1 \simeq_i \text{sphere } (\theta : \text{Space MY})$ 1)	The given bijective isometry $\Delta : \Sigma_p \rightarrow \Sigma_q$, with $q(\Delta(u) - \Delta(v)) = p(u - v)$.
show Space MX from y	Regard the same vector $y \in E$ with norm p ; this is not an additional transformation.
radialExtension (X := Space MX) (Y := Space MY) Δ (show Space MX from y)	The radial value $R_\Delta(y)$: zero at zero, and $p(y)\Delta(y/p(y))$ otherwise.
show Fin (m + 1) $\rightarrow \mathbb{R}$ from radialExtension (X := Space MX) (Y := Space MY) Δ (show Space MX from y)	Read that same value back in the fixed reference coordinates. The entire function under the derivative is $y \mapsto R_\Delta(y)$.
ContinuousLinearMap.det (fderiv $\mathbb{R}$ (fun y : Fin (m + 1) $\rightarrow \mathbb{R}$ => (show Fin (m + 1) $\rightarrow \mathbb{R}$ from radialExtension (X := Space MX) (Y := Space MY) Δ (show Space MX from y))) x)	The determinant $\det DR_\Delta(x)$ of the derivative of this whole coordinate map. The surrounding absolute-value bars in the signature give $j_\Delta(x) = \det DR_\Delta(x) $.
In the source	Mathematical meaning
MX.p.ball 0 1	The open source ball $B_p = \{x : p(x) < 1\}$.
∂volume = MY.closedUnitBallVolume	The complete equality is $\int_{B_p} j_\Delta d\lambda = \lambda(\overline{B_q})$, where $\overline{B_q} = \{y : q(y) \leq 1\}$. The cited volume definition takes the real value of its finite measure. The same reference Lebesgue measure is used throughout.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 17–21:

```

noncomputable section
open Set Metric Function MeasureTheory
open scoped NNReal ENNReal
namespace MathlibAnnex.Sphere
open EquivalentSeminorm

```

▼ Exact content identity

Declaration: MathlibAnnex.Sphere.radialExtension_integral_abs_det

Accepted content SHA-256: 56ec5070e1442d28fe4a6cb353623cd07932f03dbb626fb7c9a61afa9531c6a0

Accepted source guide SHA-256: e5c0746e9e0e08f7b2c08e6d46428030523d68af142dee003519e02cdfcc7579

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73